

EXP7 Fourier flatness sweep

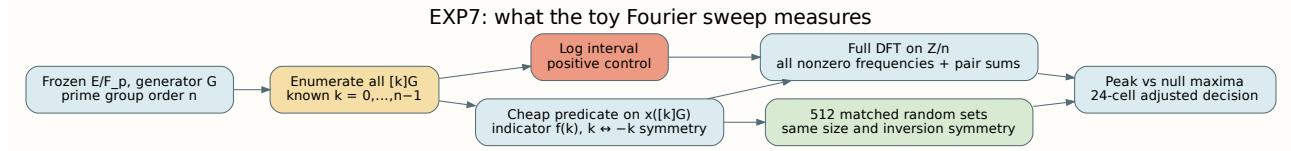
EXP7: Fourier flatness sweep on toy prime curves

Run date: 2026-10-08 (UTC; see the [date erratum](#)). **Status:** measured toy diagnostic; no adjusted candidate lead. This is the first item of the proposed six-item program. Items 2–6 were not run in this work.

Question, scope, and evidence

Does a cheap predicate of a point's affine abscissa select an unusually structured subset of the cyclic elliptic-curve group when points are indexed by their discrete logarithms? The [frozen protocol](#) specifies six candidate predicates, two null controls, one log-aware positive control, four prime-order curves, 512 matched random sets per case, and the decision rule. The [Rust sweep](#) produced the complete [raw result](#); the [peak-ratio table](#) is a compact derived view. The measured source revision is `2e3e894a45ea3cfd9490349b95081073a85c6a5f`; the baseline is `30da958ed337fb3d7522d7fd691aa15e6c733a97`.

For $P=[k]G$, let $f(k)$ be 1 if P passes the predicate and 0 otherwise. The program computes every coefficient $F(j)=\sum_k f(k) \exp(-2\pi i j k/n)$, tests the maximum for $j \neq 0$, and normalizes by $\sqrt{m(1-m/n)}$ with $m=\sum_k f(k)$. It compares this maximum with 512 independent random sets of the same size. The nulls for abscissa predicates select k and $-k$ in pairs, preserving the exact symmetry induced by $x(P)=x(-P)$. The log interval uses unconstrained random sets. The program also computes the full ordered-pair sum vector by inverse transforming $F(j)^2$, stores each membership bitset, and records the eight largest coefficients. The full bitset permits exact replay of every coefficient and pair-sum count.



The pipeline diagram's editable source is [pipeline.dot](#), and its PDF copy is [pipeline.pdf](#).

Exact measured populations

All four curves are short-Weierstrass $y^2=x^3+ax+b$ over F_p with prime group order n , cofactor 1, and generator G . They were selected by the repository constructor with curve seed `20261007`; their complete point lists in log order are in [results.json](#).

Requested bits	p	a	b	n	G=(x,y)	Exact ICV1
8	179	51	41	191	(108,39)	ICV1:fp-179:-11:191:96:unk:unk:r:a52b529ce9ce
10	739	242	567	719	(559,630)	ICV1:fp-739:21:719:614:unk:unk:r:a47a74d0df47
12	2749	1936	200	2749	(1388,830)	ICV1:fp-2749:1:2749:2173:unk:unk:r:096cb4f9e068
14	10177	3690	7829	10321	(3481,6440)	ICV1:fp-10177:-143:10321:5678:unk:unk:r:b65f5c0e2b63

For each curve, enumeration verified n distinct points, every point on the curve, and $[n]G=0$; the counted enumeration used exactly n additions and zero doublings or scalar multiplications. Curve search and its costs are in the repository's `find_prime_order_curve` constructor, not these enumeration counts. The sweep requires the known log k of every toy point and therefore is not an unknown-log membership oracle.

Results

EXP7 Fourier peaks against matched nulls

Cell = observed maximum / 95th percentile of 512 same-size null maxima

	p=179 n=191	p=739 n=719	p=2749 n=2749	p=10177 n=10321
B-smooth x	0.95×	0.82×	0.83×	0.87×
CF quotients ≤ 16	0.85×	0.89×	0.77×	0.78×
Cantor digits	0.80×	0.78×	0.89×	0.96×
Hamming weight	0.98×	0.89×	0.90×	0.85×
Farey height	0.77×	0.88×	0.77×	0.78×
Legendre +++	0.88×	1.06×	0.86×	0.81×
SHA(x) control	0.74×	0.68×	0.86×	0.79×
Random-pair control	0.94×	0.89×	0.98×	0.87×
Log-interval control	2.58×	4.70×	8.44×	15.43×

 $\leq$ null 95th percentile
 $>$ null 95th percentile
 $> 2\times$ null 95th percentile

24 candidate cells; threshold $p \leq 0.05/24$. A colored cell alone is not a lead.

The plot's editable [Rust source](#), [Cargo wrapper](#), [CSV data](#), and [PDF copy](#) are committed with the raw output. Each cell is observed normalized maximum divided by the matched-null 95th percentile. A ratio above 1 is descriptive only; the preregistered lead rule is the empirical $p \leq 0.05/24 = 0.00208333$ for any of the **24 candidate cells**. None met it.

Candidate predicate	Smallest empirical p across four curves	Largest observed/null-95 ratio
x B-smooth	0.08187	0.952
Every partial quotient of x/p at most 16	0.17154	0.890
Base-3 Cantor digits of x	0.12865	0.960
Low base-2 Hamming weight of x	0.06628	0.977
Bounded Farey height of x	0.30994	0.879
Legendre $(x, x+1, x+2)=(+, +, +)$	0.01754	1.058

The Legendre cell at $p=739$, $n=719$, has $m=84$, observed peak $Z=4.00516$, matched-null 95th percentile $Z=3.78610$, and 8 of 512 null maxima at least as large, giving $(8+1)/513=0.01754$. Its Bonferroni adjusted value is $\min(1, 24 \times 0.01754)=0.42105$; it is an unadjusted outlier, not a preregistered lead. With 512 nulls the minimum possible empirical $p=1/513=0.00194932$, so the declared threshold is crossed only if **zero** nulls exceed the observed maximum. The data support the stated bounded decision, not a universal flatness conclusion.

Control	p=179	p=739	p=2749	p=10177
SHA-256 of x, peak/null-95	0.740	0.678	0.862	0.789

Control	p=179	p=739	p=2749	p=10177
Uniform inversion-pair set, peak/null-95	0.940	0.888	0.980	0.874
Log interval, peak/null-95	2.583	4.702	8.442	15.432

The log interval exceeds its null 95th percentile on all four curves, satisfying the positive-control gate. Each log interval's empirical $p=1/513$. The SHA and random-pair controls did not exceed their own matched null 95th percentiles. There are 36 observed spectra and 18,432 null spectra in the complete four-curve sweep. All case sizes, peaks, frequencies, null distributions, and bitsets are preserved in [results.json](#).

The largest inverse-FFT pair-count distance to an integer is 7.19×10^{-9} ; the largest relative Parseval error is 1.46×10^{-12} . The reported maximum pair-sum deviation is dominated for paired sets by the forced $k+(-k)=0$ count. It is retained as a complete-convolution check, but that scalar maximum alone is not an independent additive-structure claim. If the earlier $P+QES$ count is desired, it is exactly $\sum_t (f*f)(t)f(t) = (1/n)\sum_j F(j)^2F(-j)$, recoverable from the stored bitsets; that **triple** count was not separately calibrated in this run. A maximum Fourier peak can flag a pattern missed by one sum count, but neither scalar statistic is universally more sensitive.

Reproduction and validation

From the repository root, with Rust 1.99 and the checked-in lockfile:

```
cargo test --locked --release --example exp7_fourier_flatness
cargo run --locked --release --example exp7_fourier_flatness -- \
  research/prime_fourier_flatness_exp7_20261007/results.json
cargo run --locked --release --example exp7_render -- \
  research/prime_fourier_flatness_exp7_20261007
dot -Tsvg research/prime_fourier_flatness_exp7_20261007/pipeline.dot \
  -o research/prime_fourier_flatness_exp7_20261007/pipeline.svg
bash research/prime_fourier_flatness_exp7_20261007/build_report.sh
```

The measured `results.json` is **1,419,454 bytes**, SHA-256 `cf763916bff2972d47a915c1d6dabd1c2953ecde2e82ce688ba0f5adb0655067`. A date label in the frozen protocol and study-directory name says October 7; the commit and file timestamps establish the actual UTC date as October 8. The [erratum](#) preserves this correction without rewriting the premeasurement protocol. A second execution with the same code and settings produced a byte-identical JSON file. Four focused tests pass: Bluestein versus direct DFT and inverse, pair-sum convolution versus direct counts, predicate/null behavior, and exact group enumeration. An initial premeasurement test assertion incorrectly expected $n-1$ counted group additions; it was corrected to n , including the final addition returning to 0, before the code was frozen and measured. `rustfmt --check` passes. Strict Clippy reports a pre-existing unrelated lint at `src/symmetric/modes/pmac.rs:112`; targeted Clippy with `-A clippy::mismatched_bit_width_type` passes for the example.

The algorithm pays for whole-group enumeration plus 512 FFT spectra per case; no per-target DLP cost or speedup is measured. The point lists, membership bits, null seeds, and protocol in the committed files define the exact replay. This diagnostic neither establishes the user's broad Weil/Deligne statement for every algebraic predicate nor the asserted hardcore-bit implication for every cheap predicate and arbitrary Fourier character; such claims require their own hypotheses and reductions. These toy outcomes also say nothing conclusive about larger curves.

Repository graph review and next obligations

Reviewed the canonical [index-calculus scoreboard](#), [progress data](#), [docs/curves/](#), [docs/performance-gains/](#), and [figures/](#). EXP7 introduces no IC/rho ratio, new curve registry entry, verified attack performance, or curve transport edge, so no canonical panel or published copy changes. This study's graph and diagram record the new diagnostic directly. The proposed low-degree sweep, learned classifier, jet-space descent, lift-height control, and isogeny-rank control remain unimplemented items 2-6; they are not inferred from EXP7.